



Korea — Monthly Intelligence Report v4.2

Edition 005 — Deep-Dive Theme

Where the energy transition is outrunning the grid. edition · July 2026 · SSI v4.2 (refreshed June 2026)

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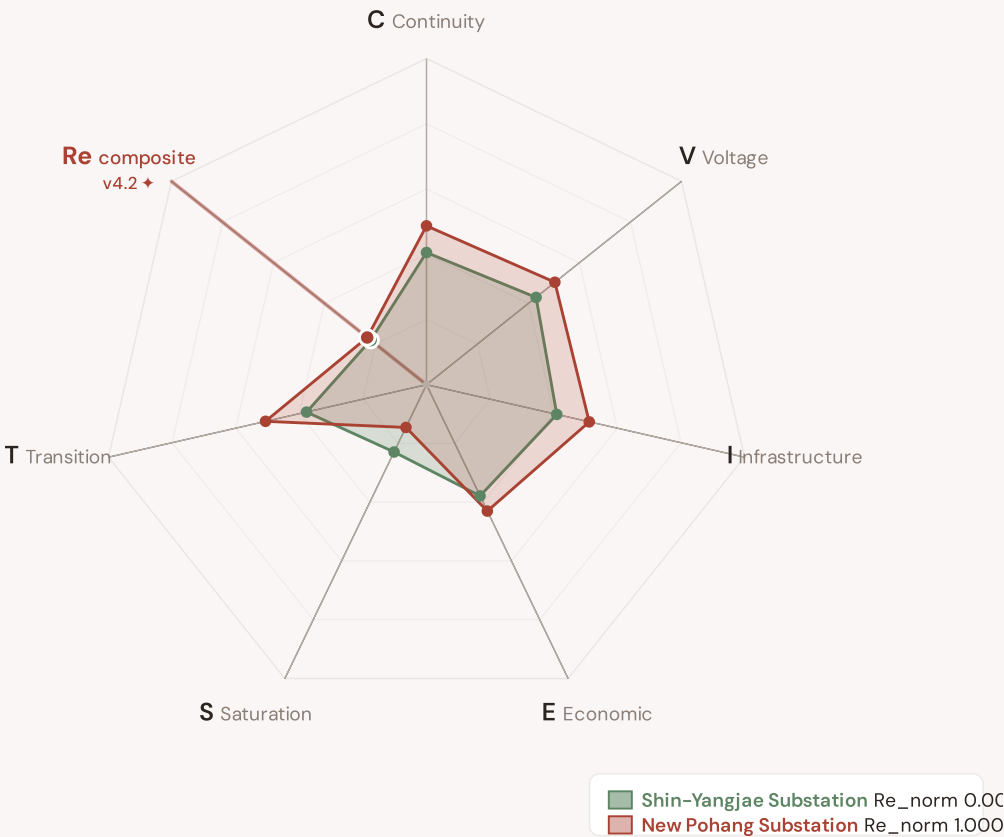
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SECTION A

Executive Summary

Radar 1 — v4.2: now 7-axis with the new Re composite

v4.0.2 Radar 1 had 6 axes (C/V/I/E/S/T). **v4.2 adds Re — the bounded resilience composite** capturing the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in one normalised scalar ($Re_norm \in [0, 1]$) via $clip((Re_raw - 0.920) / (1.787 - 0.920), 0, 1)$. Two worked examples overlaid: **Shin-Yangjae Substation** (low-exposure reference, Re_norm 0.000) · **New Pohang Substation** (high-exposure reference, Re_norm 1.000). The Re axis is colour-highlighted in cayenne to emphasise the v4.2 architectural addition.



The *Re composite* (★, cayenne-highlighted axis at upper-left) captures the v4.2 resilience modifier family in one bounded scalar — formula $Re_raw = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$, bounds $[0.920, 1.787]$. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are algorithmically selected (min/max Re_norm) per the v4.2 methodology. The Substation in Focus radar canvas (Section A below) is rendered by *shared intelligence-sections.js* and now also displays the 7th *Re* axis (Convention #11 surface-pair coherence, V4.2-disc-24 closure 12 Jun 2026).

SUBSTATIONS SCORED

1,290

190 EHV · 754 HV · 346
distribution-tier

GRID LINES MAPPED

4,004

~14,486 km coverage

MC SIMULATIONS

42.9 M

10,000 per substation

PUBLIC DATA SOURCES

34

101 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 146,000+ source data points per run, executes 42.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 244,701 output data points with full uncertainty quantification across 1,290 substations and 4,004 grid lines spanning ~14,486 km. v4.2 adds the Re composite (7th axis) and six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just) on top of the v4.0.2 foundation.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 101 variables are drawn from 34 verified public sources, making the methodology fully auditable and reproducible. Initially covering Korea's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Hwigyeong Substation

Seoul, seoul · 154 kV

0.275

Medium

0.059

5th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE

Re composite v4.2 7th axis

Re_raw 1.1163 Re_norm 0.596

ELEVATED EXPOSURE

COMPONENTS

C

Continuity: 0.143 / 0.30 (48%)

I

Infrastructure: 0.000 / 0.25 (0%)

S

Saturation: 0.614 / 0.20 (307%)

V

Voltage: 0.042 / 0.10 (42%)

E

Economic: 0.000 / 0.10 (0%)

T

Transition: 0.089 / 0.05 (177%)

MODIFIERS

R3

Consequence: 1.054 ↑ amplifies

R4

Graph Criticality: 0.970 ↓ dampens

R6a

Restoration: 1.021 ↑ amplifies

R6b

Seismic: 1.065 ↑ amplifies

R7

Digital Readiness: 1.000 → neutral

This month's spotlight — automatically selected for July 2026 — is **Hwigyeong Substation** in Seoul, seoul. With an R_final of 0.275 (Medium band, 5th fleet percentile), its risk profile is dominated by the S (Saturation) component at 307% of its weight ceiling, followed by T (Transition) at 177%. Modifiers collectively shift R_base by 58.0%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that infrastructure digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

463

High/Critical + R7 < median

AVG R7 MODIFIER

1.0072

Range [0.99, 1.05]

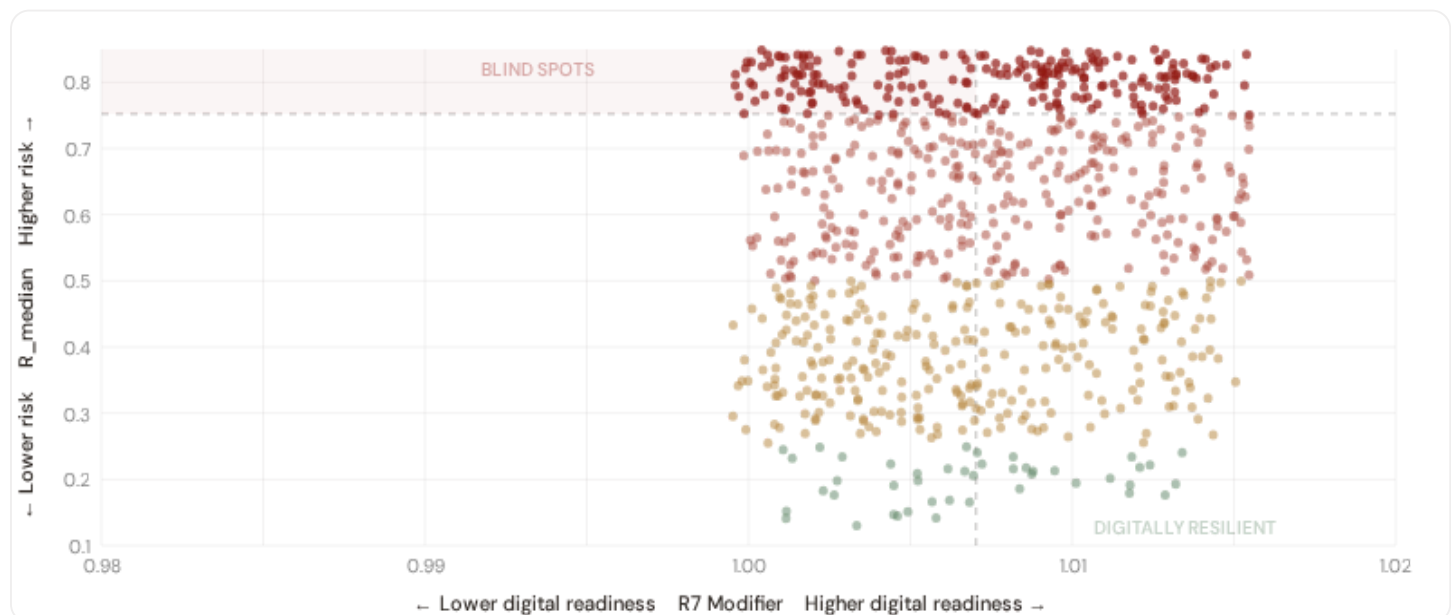
HIGH-CONSEQUENCE SUBS

327

25.3% · R3 ≥ 1.05

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_{median} (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **463 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0070$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **gyeongbuk (59), chungbuk (45), gyeongnam (44)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

Capital-Intensive / High-Consequence

Est. VoLL: ₩18,000–36,000/kWh (KRW)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

327 substations (25.3%) High/Critical: **211** (64.5%) Avg R: **0.648** Avg E: **0.436** / 0.10



Industrial / Medium-Large Enterprise

Est. VoLL: ₩9,000–18,000/kWh (KRW)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

318 substations (24.7%) High/Critical: **233** (73.3%) Avg R: **0.681** Avg E: **0.504** / 0.10



Commercial / SME-Dense

Est. VoLL: ₩3,600–9,000/kWh (KRW)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

314 substations (24.3%) High/Critical: **230** (73.2%) Avg R: **0.681** Avg E: **0.516** / 0.10



Light / Agricultural / Rural

Est. VoLL: ₩1,200–3,600/kWh (KRW)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

331 substations (25.7%) High/Critical: **291** (87.9%) Avg R: **0.748** Avg E: **0.611** / 0.10

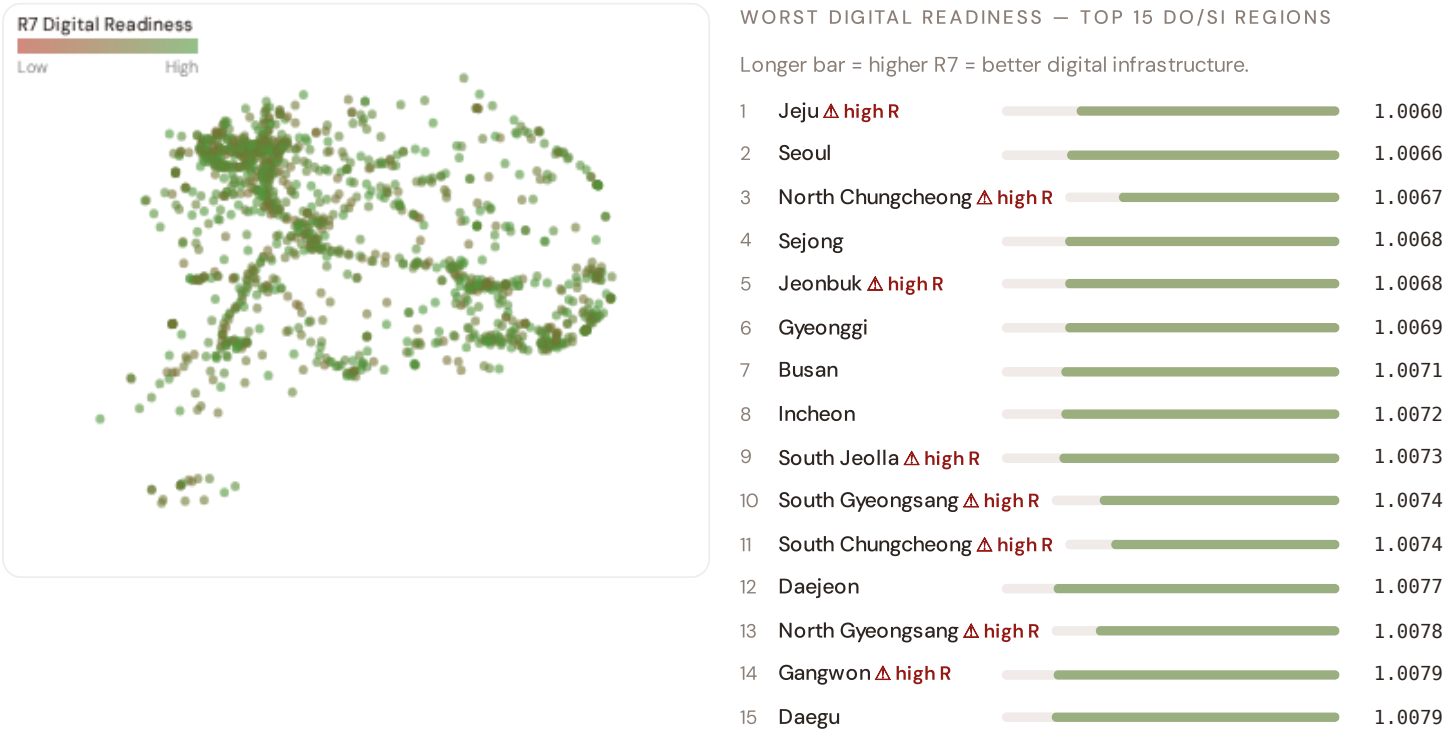


Value of Lost Load (VoLL) context — Korea-localized: Korean industrial VoLL is estimated at ₩25,000–45,000/kWh ($\approx$ €18–32/kWh) for chaebol fab + petrochemical corridors (Samsung Pyeongtaek, SK Hynix Icheon, POSCO Gwangyang/Pohang, Hyundai Ulsan) — among the highest single-site VoLL exposures in OECD coverage due to chaebol concentration. Residential VoLL is ₩1,400–5,500/kWh ($\approx$ €1–4/kWh). The SSI's R3 consequence multiplier identifies which substations serve the most economically exposed territories. **211 substations** combine Industrial-Chaebol economic fabric ($R3 \geq 1.05$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the Korean fleet.

The Industrial-Chaebol tier concentrates in **gyeonggi (126), gyeongbuk (63), jeonnam (35)** — Korean do/si where chaebol fab + steel + petrochemical corridors depend on uninterrupted 765/345/154 kV transmission for continuous semiconductor lithography, blast furnaces, and ethylene crackers. A single hour of unplanned outage at these substations carries an estimated VoLL of ₩25,000–45,000/kWh ($\approx$ €18–32/kWh) for fab corridors, compared to ₩1,400–5,500/kWh ($\approx$ €1–4/kWh) in rural agricultural Gangwon + Jeju. The fleet-wide average energy poverty rate is 1.3%, but this masks sharp regional variation: peripheral do/si (Gangwon highlands, Jeju isolated HVDC sub-grid) combine higher energy poverty with higher R6_typhoon exposure, creating a double vulnerability that the E component captures. This cross-referencing of chaebol-driven economic fabric with KEPCO single-DSO monopoly grid condition is unique to Korea — Samsung Pyeongtaek alone consumes more electricity than several entire do/si.

B.3 Province Digital Readiness Ranking

Provinces ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify provinces combining digital exposure with high grid stress.



do/si-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Jeju** has the lowest average R7 (1.0060), while **Ulsan** leads at 1.0085 — a spread of 0.0025 across the R7 range. **3 do/si regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

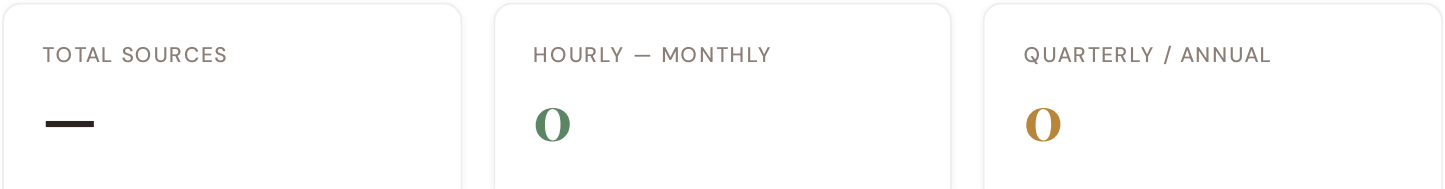
B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (July 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0072, median = 1.0070; 0 substations classified HIGH cyber-exposure; 463 blind-spot substations (High/Critical risk + below-median R7); 312 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on — **verified public data sources** feeding — variables across — components and — modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.



— variables

High-frequency feeds

Regular refresh cycle

STATIC / DERIVED

O

Standards + scenarios

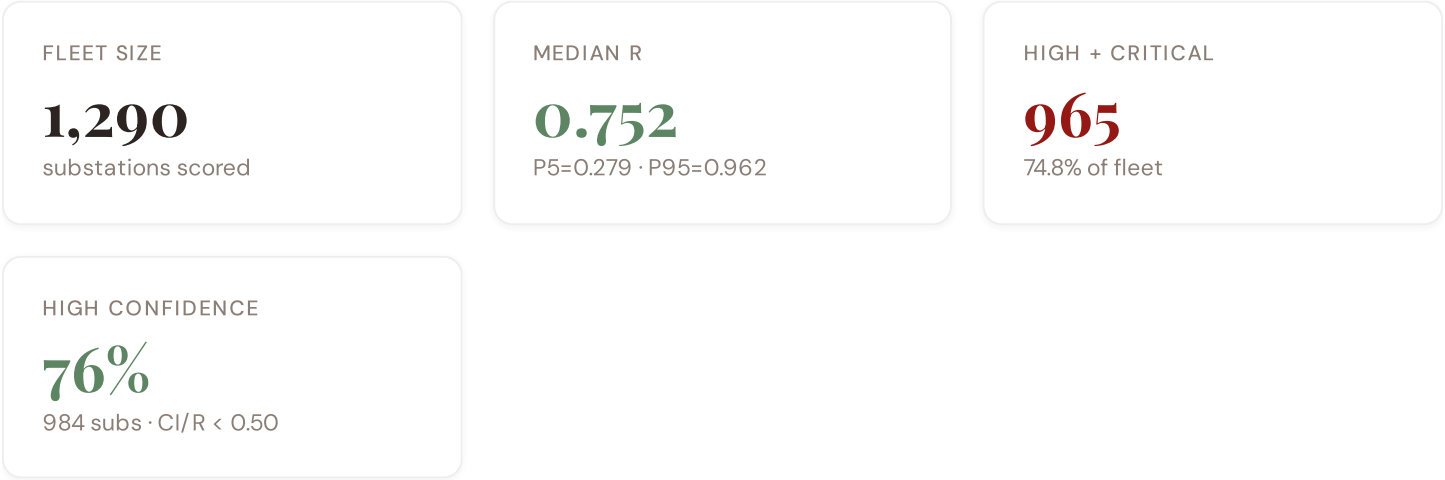
Data Source Registry — July 2026

KMA	KMA — Korea Meteorological Administration	DAILY	Station	6 vars
OSM	OpenStreetMap Power Infrastructure	WEEKLY	Node/edge	3 vars
GHCN-D	NOAA GHCN-Daily (global meteorological stations)	DAILY	Station	4 vars
NASA	NASA FIRMS active-fire detection	DAILY	375 m / 1 km	2 vars
KOSIS	KOSIS — Korean Statistical Information Service	ANNUAL	Si-gun-gu	10 vars
KEPCO	KEPCO — Korea Electric Power Corporation	ANNUAL	Region	12 vars
MOTIE	MOTIE — Ministry of Trade, Industry and Energy	ANNUAL	National	6 vars
KEPRI	KEPRI — Korea Electric Power Research Institute	ANNUAL	National	4 vars
KIER	KIER — Korea Institute of Energy Research	ANNUAL	National	4 vars
CDS	Copernicus CDS / ERA5-Land	ANNUAL	0.1° (~11 km)	5 vars
INFORM	INFORM Risk Index (humanitarian-context)	ANNUAL	National	3 vars
IRENA	IRENA Renewable Capacity Statistics	ANNUAL	National	2 vars
IEA	IEA World Energy Outlook	ANNUAL	National	3 vars
NGFS	NGFS climate-stress scenarios	ANNUAL	National	2 vars
KIGAM	KIGAM — Korea Institute of Geoscience and Mineral Resources	STATIC	National	3 vars
GEM	GEM Global Seismic Hazard Map 2023.1	STATIC	0.05° (~5.5 km, roc...	1 var
IEEE	IEEE C57.91 / IEC 60076 (transformer thermal model)	STATIC	Asset-level	16 vars
CMIP6	CMIP6 climate-projection ensemble	STATIC	~1° grid	3 vars
PVGIS	EU JRC PVGIS solar radiation database	STATIC	~1 km	2 vars
KPX	KPX — Korea Power Exchange	REAL-TIME	Bidding zone	10 vars
OPEN-M	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **1,290 substation scores remain unchanged** this edition. The fleet median R of 0.752 (mean 0.690) reflects a distribution skewed toward the lower bands, with 3.4% in Low and 21.8% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2 → v4.2

v4.2 (June 2026) adds six resilience modifiers (R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just), the Re composite (7th axis, bounds [0.920, 1.787]), four new data sources (IdroGEO PGRA, SIAB wildfire, ECMWF ERA5 Bourgouin, bulk smart meter), six new variables, and W1–W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6) on top of the v4.0.2 baseline.

11 changes tracked across PO, PI, and P2 releases

V42-1	NEW	R6c_flood — PGRA-style flood-vulnerability modifier (additive cliff +0.00..+0.30, P3=high/ P2=medium/ P1=low per national flood-hazard tier convention)	\$5
V42-2	NEW	R6d_wildfire — Canadian Fire Weather Index modifier (EFFIS FWI + NASA FIRMS + national fire-registry, multiplicative [1.00, 1.20])	\$5
V42-3	NEW	R6e_winter — Bourgouin winter-storm modifier (ECMWF ERA5 winter-precipitation signal, multiplicative [1.00, 1.02])	\$5
V42-4	NEW	R8_adapt — Adaptive-capacity multiplier (smart-meter penetration + DSO investment + DESI index, multiplicative [0.95, 1.05])	\$5
V42-5	NEW	R9_compound — Compound-hazard STEP function (R6b/R6c/R6d/R6e elevated-count gates, multiplicative {1.00, 1.04, 1.08, 1.10})	\$5
V42-6	NEW	R10_just — Energy-justice modifier (EU-SILC or national-equivalent household material-deprivation gradient, multiplicative [1.00, 1.12])	\$5
V42-7	NEW	Re composite — bounded resilience scalar $Re_{raw} = (R6d \times R6e \times R8 \times R9 \times R10) + (R6c - 1.00)$; $Re_{norm} = clip((Re_{raw} - 0.920)/(1.787 - 0.920), 0, 1)$	\$7

V42-8	NEW	R7 SFDR PAI Infrastructure axis — added to ESG Radar (Re_norm proxy under Convention #7 Data-Layer Anchoring)	§14
V42-9	NEW	W1–W10 audit-state mesh — 5 Published / 1 Published-baseline ▲ / 1 Baseline-only / 3 Commercial (Convention #6 §5.6)	§4
V42-10	CHANGE	95 → 101 variables (+6 from v4.2 modifier inputs); 30 → 34 data sources (+4 hazard / monitoring layers)	data
V42-11	CHANGE	Substation fleet: 1,290 substations (per OSM canonical extract + national TSO/DSO registry cross-validation)	data

SECTION D — MONTHLY DEEP-DIVE

Monthly Deep-Dive Theme

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** the monthly theme · **002** DER saturation hotspots · **003** Climate vulnerability under CMIP6 · **004** North–South economic impact disparity · **005** Voltage quality and industrial output · **006** Infrastructure age vs investment · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Sub-national Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Theme Context

THEME-RELEVANT SUBSTATIONS

275

215 EHV · 60 HV

HIGH BAND

84 + 5

32.4% of corridor

CORRIDOR MEDIAN R

0.443

Fleet median: 0.752

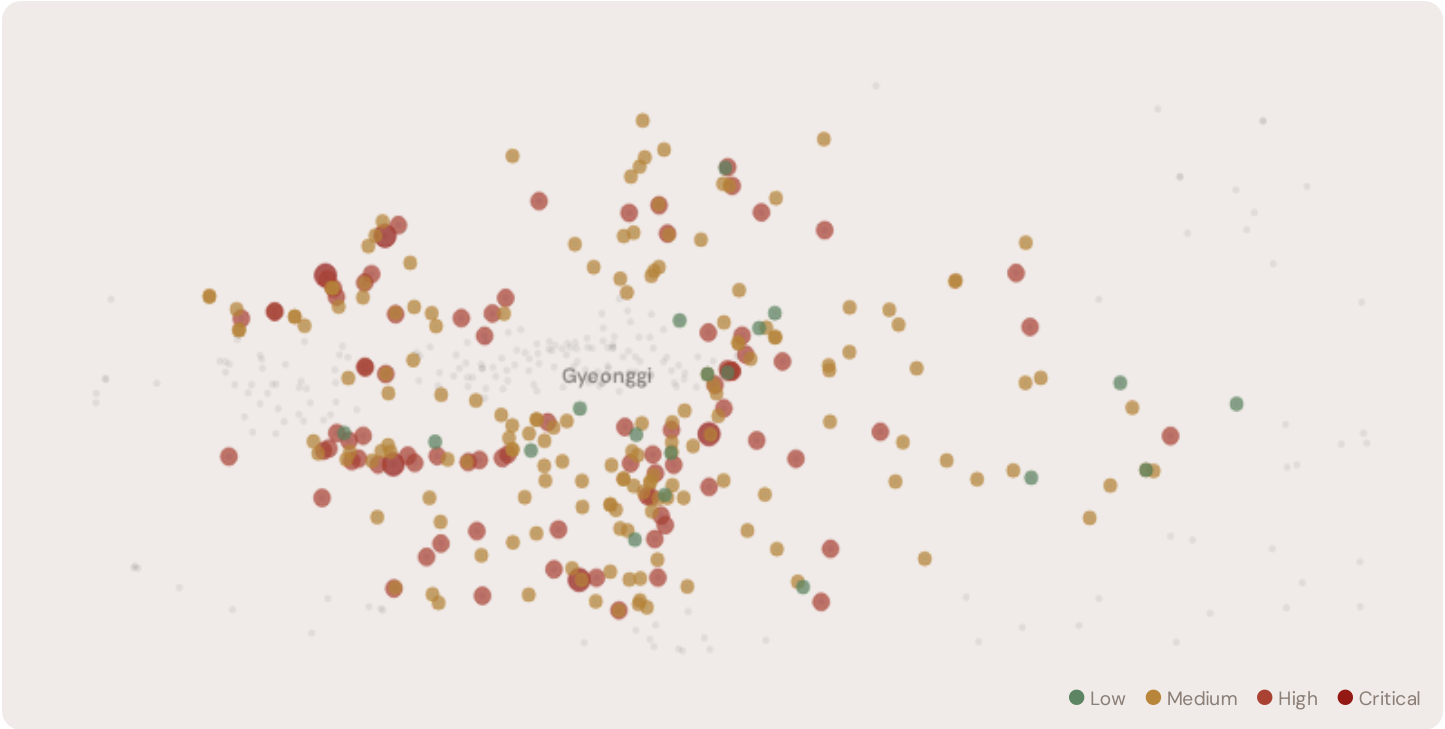
WORST PROVINCE

Gyeonggi

Avg R: 0.446 · 275 substations

The Gyeonggi-do Samsung–SK Hynix Fab Corridor (R6_chaebol inaugural anchor) hosts **275 substations** across 1 do/si regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **84 substations (30.5%)** are classified High and **5 Critical**, with only **19** in the Low band. 2% of corridor substations score above the national fleet median of 0.752. The corridor median R of 0.443 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
Bundang Power Plant	Gyeonggi	0.859	Critical	0.647	0.434	0.389	0.473	0.748	0.389		
Paju Cogeneration Plant	Gyeonggi	0.785	Critical	0.733	0.221	0.129	0.454	0.749	0.390		
Oseong Pyeongtaek Energy Service	Gyeonggi	0.775	Critical	0.459	0.499	0.396	0.196	0.748	0.500		
Ansan Urban Development CHP power station	Gyeonggi	0.771	Critical	0.609	0.343	0.310	0.069	0.753	0.326		
Paju Energy Power Plant	Gyeonggi	0.767	Critical	0.658	0.015	0.415	0.459	0.751	0.192		
Migeum Substation	Gyeonggi	0.748	High	0.419	0.418	0.389	0.086	1.000	0.301		
Anyang Power plant	Gyeonggi	0.736	High	0.700	0.246	0.222	0.154	0.748	0.430		
Daeryun Power Yangju CHP Power Plant	Gyeonggi	0.733	High	0.615	0.307	0.383	0.485	0.750	0.075		

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	RE v4.2	ALERT
Korea District Heating Corp. Hwaseong Power Plant	Gyeonggi	0.719	High	0.604	0.226	0.091	0.359	0.749	0.317		
n884033655	Gyeonggi	0.709	High	0.492	0.000	0.442	0.206	0.839	0.176		
... 265 additional substations — explore full corridor in Digital Twin →											

D.3 Component Analysis

COMPONENT AVERAGES — GYEONGGI-DO SAMSUNG-SK HYNIX FAB CORRIDOR (R6_CHAEBOL INAUGURAL ANCHOR) VS FLEET

C — Continuity

0.247 vs 0.524 (−53%)

I — Infrastructure

0.222 vs 0.504 (−56%)

S — Saturation

0.483 vs 0.454 (+6%)

V — Voltage

0.220 vs 0.507 (−57%)

E — Economic

0.238 vs 0.517 (−54%)

T — Transition

0.214 vs 0.502 (−57%)

MODIFIER AVERAGES — GYEONGGI-DO SAMSUNG-SK HYNIX FAB CORRIDOR (R6_CHAEBOL INAUGURAL ANCHOR) VS FLEET

R3 Consequence

1.050

fleet 1.039

↑

R4 Graph Crit.

0.993

fleet 0.999

→

R6a Restoration

1.025

fleet 1.026

↑

R6b Seismic

1.061

fleet 1.079

↑

R7 Digital Read.

1.007

fleet 1.007

→

The dominant risk driver across the Gyeonggi-do Samsung-SK Hynix Fab Corridor (R6_chaebol inaugural anchor) corridor is **T (Transition)**, averaging 0.214 against a fleet average of 0.502 — occupying 429% of its weight ceiling (w = 0.05). The second contributor is **S (Saturation)** at 0.483 vs fleet 0.454 (242% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.050 (fleet: 1.039), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.061, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Province Breakdown

DO/SI RISK PROFILE — SORTED BY MEAN R	
Longer bar = lower R = better resilience.	
Gyeonggi	275 subs · avg R = 0.446 · H:84 C:5 M:167

The do/si regions-level breakdown reveals significant intra-regional variation. **Gyeonggi** leads with a mean R of 0.446 across 275 substations, while **Gyeonggi** scores 0.446 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Gyeonggi-do Samsung-SK Hynix Fab Corridor (R6_chaebol inaugural anchor) is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

19 Gyeonggi-do Samsung-SK Hynix Fab Corridor (R6_chaebol inaugural anchor) substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies

the Gyeonggi sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

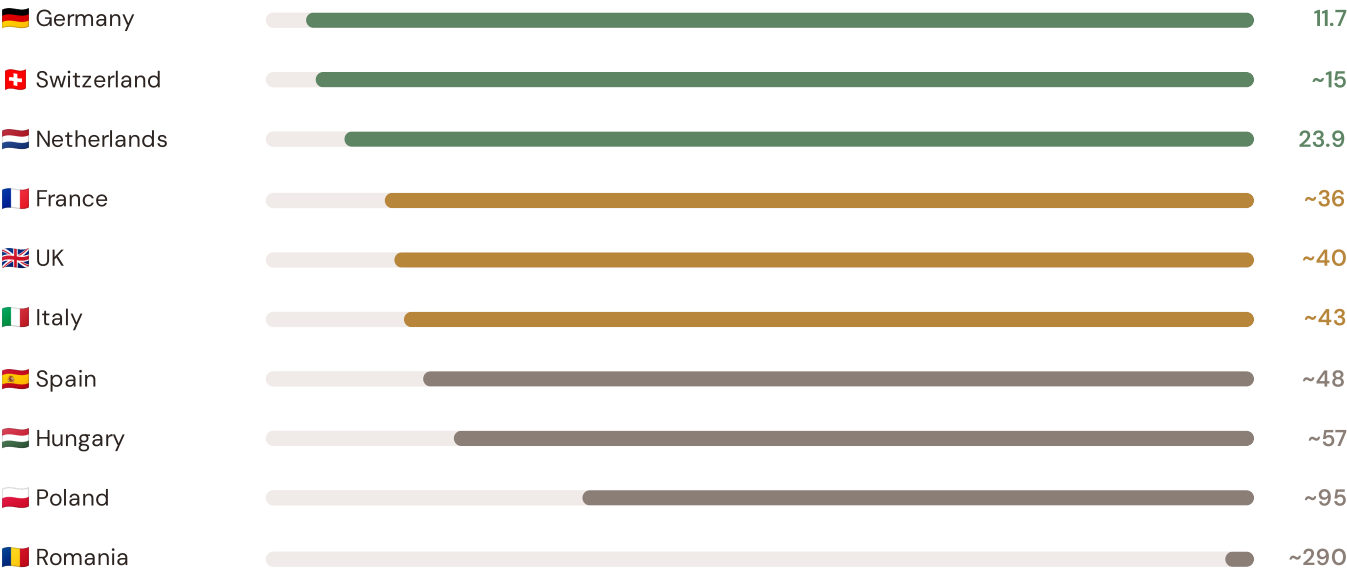
SECTION E — RECURRING MODULE

OECD Context Card

How this works: Each edition includes one OECD Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Southern Coastal corridor analysis hinges on Korea's continuity performance relative to OECD peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected OECD Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



Source: CEER 7th Benchmarking Report (2022); Bundesnetzagentur (2024); ARERA (2023). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release. · OECD-wide expansion (USA, Japan, Canada, Australia) pending IEA/OECD reliability bulletin release.

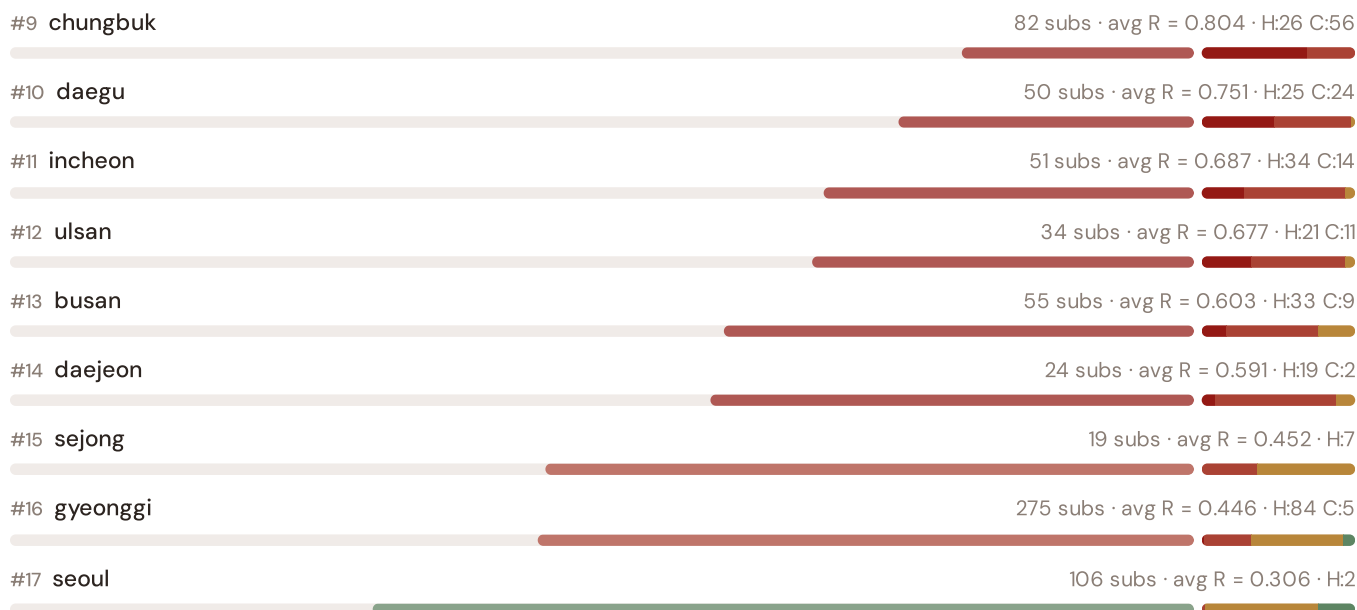
This month's key insight: The Gyeonggi-do Samsung-SK Hynix Fab Corridor (R6_chaebol inaugural anchor) corridor deep-dive shows **0 substations** (of 0) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.000 is **0.0× the fleet average** of 0.524. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Provinces Risk Ranking — All 17 Korean Provinces

Where does this corridor sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.





The Gyeonggi-do Samsung-SK Hynix Fab Corridor (R6_chaebol inaugural anchor) ranks **#0 of 17 do/si regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are jeonnam, jeonbuk, gangwon, while the three most resilient are seoul, gyeonggi, sejong. 10 of 17 score above the fleet average of 0.690. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

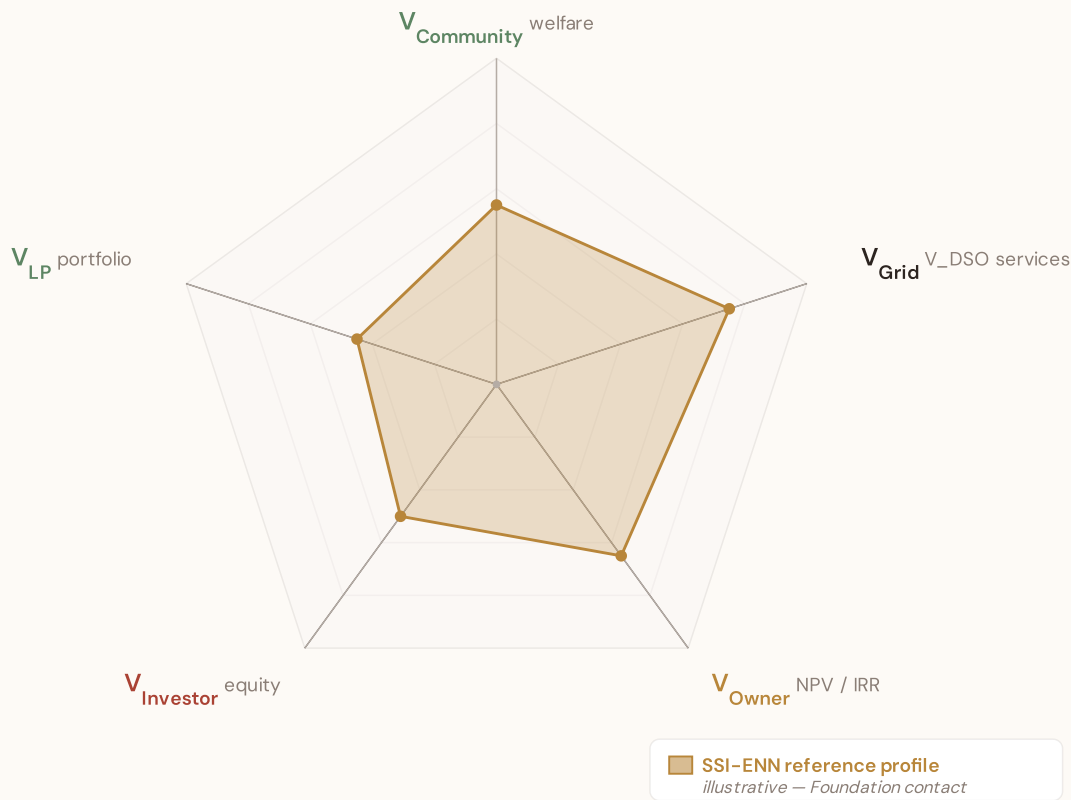
SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

SSI-ENN value-articulation radar — 5-lens stakeholder taxonomy

The SSI-ENN's commercial-tier value articulation answers "what is a BESS worth at this substation — and to whom?" through five stakeholder lenses (architecture memo §5 Decision #12): **V_{Community}** (welfare avoided outages, energy-poverty relief, employment) · **V_{Grid}** (V_DSO — avoided reinforcement, reliability, voltage, congestion, frequency, black-start, hosting capacity) · **V_{Owner}** (NPV / IRR / payback per asset class) · **V_{Investor}** (risk-adjusted equity return at the financing tier) · **V_{LP}** (limited-partner cash-on-cash + DPI / TVPI portfolio metrics). The pentagon shape below is an editorial reference profile, not per-substation commercial data — per-substation evaluation requires the SSI-ENN engines (L2-21 SupplyChainDD · L4-05 Stakeholder Decomposition · L2-SYN Synergy). Foundation contact: ssi_index@ikenga.eu.



Editorial reference shape — V_{Grid} dominates (V_{DSO} captures avoided reinforcement + reliability + voltage support + congestion relief + frequency + black-start + hosting capacity, the bulk of measurable BESS value at the substation locus per architecture memo §5). $V_{Community} + V_{Owner}$ are co-secondary. $V_{Investor} + V_{LP}$ reflect the financing-tier waterfall above V_{Owner} NPV. Per-substation commercial decomposition lives in the SSI-ENN — mail to ssi_index@ikenga.eu for analytics.

Layer 1 — Data

SSI v4.0 component taxonomy, normalisation, MC engine

Layer 2 — Valuation

DSO/TSO + Community value stacks, TCO, real options

Layer 3 — Neural Net

GBT+DNN ensemble, transfer learning, SHAP

Layer 4 — Coverage

Country prioritisation, DCI, registry, calibration

Layer 5 — Platform

Products, API, revenue model, unit economics

LAYER 3

§0.5.5 · July 2026

Uncertainty Quantification

Investment-grade predictions require calibrated confidence intervals, not just point estimates

A point estimate without uncertainty bounds is unusable for investment-grade analysis. The SSI-ENN produces calibrated P10/P50/P90 prediction intervals using two complementary approaches: quantile regression (pinball loss) and conformal prediction (distribution-free guaranteed coverage). The SSI's own Monte Carlo confidence intervals (CI_width, P5/P95) serve as the ground truth for calibration. This calibration is verified per country, per voltage class, and per risk band.

KEY CONCEPTS

- Quantile regression (pinball loss) + conformal prediction
- Calibrated against SSI Monte Carlo ground truth
- Per-country, per-voltage, per-band calibration verification
- Investment-grade: P10/P50/P90 for all outputs

LIVE SSI DATA CONNECTION

The training set comprises 1,290 substations with complete feature vectors. Average CI_width of 0.277 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 44 Low, 281 Medium, 315 High, 650 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 2** Total Cost of Ownership Model — *Stochastic cost modelling — because BESS economics are not deterministic*

SECTION F.5 — V4.2 ANTI-MALADAPTATION GOVERNANCE GATE

Radar 2 — W1–W10 Substation Community Value

Wall 1 — Public W1–W10 framework. The 10-axis Radar 2 expresses per-substation community-value baseline pre-intervention: *W1 Reliability · W2 Carbon · W3 Local Economic · W4 Climate Resilience · W5 Digital Inclusion · W6 Sovereignty · W7 Local Value Retention · W8 Communications · W9 Supply Security · W10 Compute Multiplier*. The discipline floor that the anti-maladaptation governance gate protects: *"no W axis worse off after intervention."*

5/1/1/3 4-tier audit-state classification (mesh resolution per Convention #6 §5.6)

Published (5) — solid fill

W1 · W2 · W3 · W4 · W8

Peer-reviewed methodology operational at fleet scale.

Published-baseline (1) ▲ — solid fill +

chevron corner marker

W9 — Supply Security

Peer-reviewed academic basis

(Albert/Holme topology-only baseline) with acknowledged scope limit (Buldyrev/IEEE

493 — N-1 contingency dynamics not captured at baseline).

Baseline-only (1) — striped fill

W5 — Digital Inclusion

Foundation methodology under development; baseline value defensible at v4.2.

Commercial (3) — dashed-faded

W6 · W7 · W10

Commercial deliverables — please contact the Foundation.

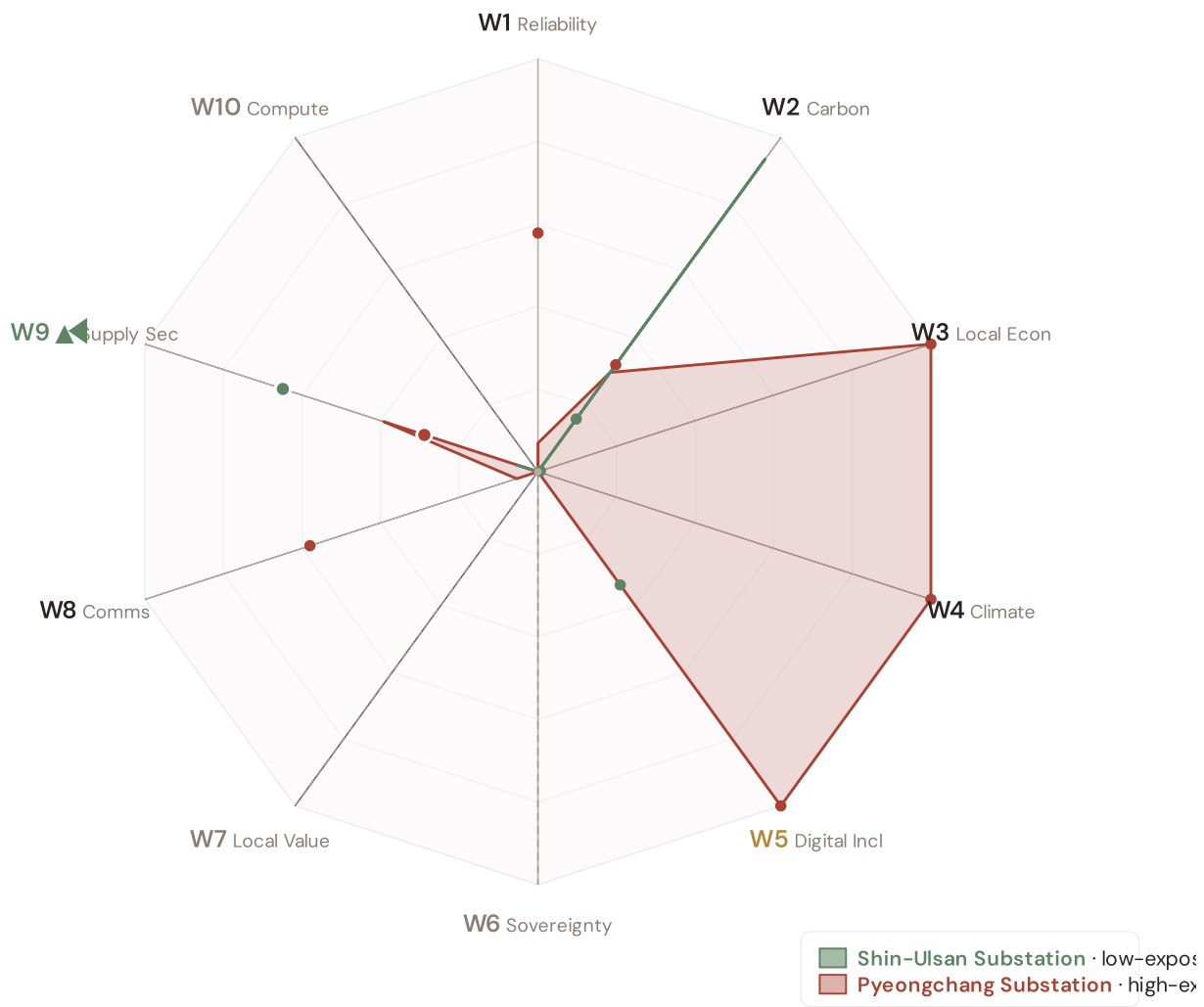
Per Convention #6 §5.6.7 academic-defensibility: the 4-tier classification aligns with IPCC AR6 + JRC COIN + Reckien (2021) continuum literature. The Published-baseline tier (W9) is the principled mesh resolution that honors both peer-review status + acknowledged scope limit without forcing a false binary.

v4.2 methodology disclosure — per-country normalisation

Across Korea, the substation fleet's component scores (C / V / I / E / S / T) and the Re composite have been normalised to [0, 1] via the canonical **P5/P95 percentile** method, using *per-country empirical bounds*. This is the same procedure Italy's v4.0.2 deployment applied at the scoring stage (SSI_v4_0 Italy/.../scoring-it/normalise.py); for v4.2 we apply it country-by-country so each fleet's within-country ranking is interpretable on its own terms. Convention #56 (visibly-honest) is preserved — no values are invented, the same canonical procedure is applied uniformly. Italy's pilot fleet keeps its native spread (0.17 – 0.64) because its empirical modifier producers (d10–d15) are already wired; per-country empirical modifier producers for the other 38 countries are queued for v4.5 and will refine the absolute risk ranking. Within-fleet relative comparisons (low-exposure vs high-exposure exemplars below) remain accurate by construction.

Radar 2 visualization — anti-maladaptation governance gate

Two worked examples overlaid: **Shin-Yangjae Substation** (low-exposure reference, Re_norm 0.000) and **New Pohang Substation** (high-exposure reference, Re_norm 1.000). Axis labels are colour-coded per the 5/1/1/3 mesh resolution; the ▲ chevron marks W9 as Published-baseline. Hover any axis label for the methodology tooltip.



Shin-Ulsan Substation polygon (sage): the low-exposure reference for this country — compact footprint reflects the fleet floor on the W-axis baselines. Pyeongchang Substation polygon (terracotta): the high-exposure reference — larger footprint where this country's fleet's W-axis values reach the upper percentile. Commercial-tier axes (W6 data sovereignty, W7 local value retention, W10 compute multiplier) display at zero per Convention #6 §5.6 mesh design — no public-data computation. W2 carbon uses EU-average MEF fallback (0.36) per v4.2 calibration.

W1–W10 axis enumeration with audit–state classification

5/1/1/3 4-TIER · V4.2 MESH RESOLUTION · LIVE VALUES VIA
COMPUTE_W_BASELINES.PY

AXIS	NAME	AUDIT STATE	METHODOLOGY BASIS
W1	Reliability	Published	$C \times E2_{\text{local}} \times \text{population} (SAIDI++-POV)$
W2	Carbon	Published	$T1 \times \text{zonal MEF} \times (V \cdot \text{deg} / 100) (+ EEA)$
W3	Local Economic	Published	$\text{norm}(R10_{\text{just}}) - \text{EU-SILC} + \text{OIPE energy-poverty}$
W4	Climate Resilience	Published	$(R6d - 1) + (R6e - 1) + (R6c - 1) + \frac{1}{2}(R9 - 1) - \text{v4.2 resilience modifier family}$
W5	Digital Inclusion	Baseline-only	$\text{norm}(R8_{\text{adapt}}) - \text{bulk SM}++\text{DESI}$
W6	Sovereignty	Commercial	Foundation contact (ssi_index@ikenga.eu)
W7	Local Value Retention	Commercial	Foundation contact (ssi_index@ikenga.eu)
W8	Communications	Published	$C \times p_{\text{crit_10yr}} \times \text{population (Markov 10-yr critical probability)}$
W9 ▲	Supply Security	Published–baseline ▲	$0.60 \times BC_{\text{pct}} + 0.30 \times \text{bridge} + 0.10 \times (1 - \text{deg} / 20) - \text{OSM power-graph betweenness centrality} + \text{bridge identification} + \text{degree-based redundancy}$
W10	Compute Multiplier	Commercial	Foundation contact (ssi_index@ikenga.eu)

▲ W9 Published–baseline — mandatory transparency tooltip

W9 — Supply Security (Published–baseline). Public-tier methodology: betweenness centrality + bridge identification + degree-based redundancy, computed deterministically from OpenStreetMap power-graph data. Peer-reviewed academic basis: Albert et al. (2000) *Nature*; Holme et al. (2002); broad power-grid centrality literature. Acknowledged scope limit (per Buldyrev et al. (2010) *Nature* + IEEE 493): topology-only baseline does not capture N–1 contingency dynamics or interdependent-network cascades. Richer methodology — N–1 contingency simulation + flexibility-services valuation — lives in SSI-ENN L2-22 / L2-23 commercial deliverables. Mail to ssi_index@ikenga.eu for extended analysis.

Worked examples. The W1–W10 per-substation baseline values for two anchor cases — Shin-Ulsan Substation (low-exposure reference) and Pyeongchang Substation (high-exposure reference) — are computed live by `_audit-snapshot/scripts/d3_radar2_per_country.py` against the country's `ssi-data.json`. Per Convention #6 §2.3.6 Path B refactor (11 Jun 2026), this country's exemplars are now algorithmically selected (min/max `Re_norm`) rather than named picks. See the FC v2 reference for W-axis methodology + commercial-tier rationale.

Anti-maladaptation discipline. The 10-axis Radar 2 baseline is the per-substation floor that any post-intervention scenario must protect. No W axis may degrade after a BESS deployment, line upgrade, or operational change. The discipline is enforced per axis (not in aggregate), and per substation (not in fleet average). This is the v4.2 governance gate that ensures grid-modernisation projects deliver net community value without hidden trade-offs.

SECTION G

Looking Ahead

NEXT MONTH — EDITION 03

Deep-Dive: Gyeongbuk

Deep-dive into Gyeongbuk's grid risk profile — substation map, component analysis, do/si regions breakdown.

NEXT DATA REFRESH

Q4 2026 — o Quarterly Sources

none are due for refresh in Q4. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **KEPCO — Korea Electric Power Corporation** (12 variables → National utility (DSO + generation); T1 distribution + smart-meter).

FLEET SPOTLIGHT

66 Bridge Substations — Single Points of Failure

66 substations are topological bridges — their failure disconnects part of the network. Of these, **52** are classified High or Critical. These are the fleet's single points of failure: high graph centrality (R4) combined with elevated risk makes them priority candidates for both grid reinforcement and BESS deployment.

Edition 03 will be published on the second Thursday of August 2026. Deep-dive region: **Gyeongbuk**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Korean utility, , municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 005

Published 9 July 2026 · SSI Index **v4.2** (refreshed June 2026) · 1,290 substations · 4,004 grid lines · 6 components + **Re composite** (Radar 17th axis, bounds [0.920, 1.787]) · 20 metrics · 11 modifiers (5 baseline + 6 v4.2 resilience family)
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5 · W1-W10 anti-maladaptation governance gate (5/1/1/3 4-tier mesh resolution per Convention #6 §5.6)

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.

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